

Verify whether following series are convergent or not

1) $\frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{n!} + \dots$

2) $\sum_{n=2}^{+\infty} \frac{\ln n}{n^2}$

3) $\sum_{n=1}^{+\infty} \frac{n^3 + 3\ln n - \ln n}{n^6 + e^n}$

4) $\sum_{n=2}^{+\infty} \frac{1}{n \ln n}$

5) $\sum_{n=2}^{+\infty} \left(\frac{1}{n} + \frac{1}{n+1} \right)$

6) $\sum_{n=2}^{+\infty} \left(\frac{1}{n} - \frac{1}{n+1} \right)$

Find the limit of the series, using ϵ - δ def

a) $\sum_{n=1}^{+\infty} n^2$

Find the value of x , which series converge:

a) $\sum_{n=1}^{+\infty} (-5)^n x^n$

b) $\sum_{n=1}^{+\infty} (x+2)^n$

Since $\sum_{n=1}^{+\infty} a_n = \lim_{N \rightarrow \infty} \sum_{n=1}^N a_n$, we can introduce $\sum_{n=1}^N a_n =: S_N$. Then $\sum_{n=1}^{+\infty} a_n$ converge

$\Leftrightarrow \exists \lim_{N \rightarrow \infty} S_N = a \Leftrightarrow \forall \epsilon > 0 \exists N_0 \in \mathbb{N}:$

$\forall N > N_0 \quad |S_N - a| < \epsilon$

• $\sum_{n=1}^{+\infty} a_n$ diverges $\Leftrightarrow \forall C > 0 \exists N_0 \in \mathbb{N}:$

$\forall N > N_0 \quad |S_N| > C$

Find limits
using ϵ - δ def.

1) $\lim_{n \rightarrow \infty} (\sqrt{n+1} - \sqrt{n})$

2) $\lim_{n \rightarrow \infty} \left(\frac{1}{27} - \frac{1}{81} + \frac{1}{243} - \dots + \frac{(-1)^n}{3^{n+1}} \right)$

Find limits:

1) $\lim_{n \rightarrow \infty} \frac{\sqrt{n}}{\sqrt{n + \sqrt{n + \sqrt{n}}}}$

Hint: $\frac{n}{2^{n-1}} = \underbrace{\frac{1}{2^{n-1}} + \dots + \frac{1}{2^{n-1}}}_{n \text{ times}}$

2) $\lim_{n \rightarrow \infty} \left(1 + \frac{2}{2} + \frac{3}{4} + \frac{4}{8} + \frac{5}{16} + \dots + \frac{n}{2^{n-1}} \right)$